
Exposing the Future

A Mathematical Framework for Detecting Data Leakage in Medical
Machine Learning

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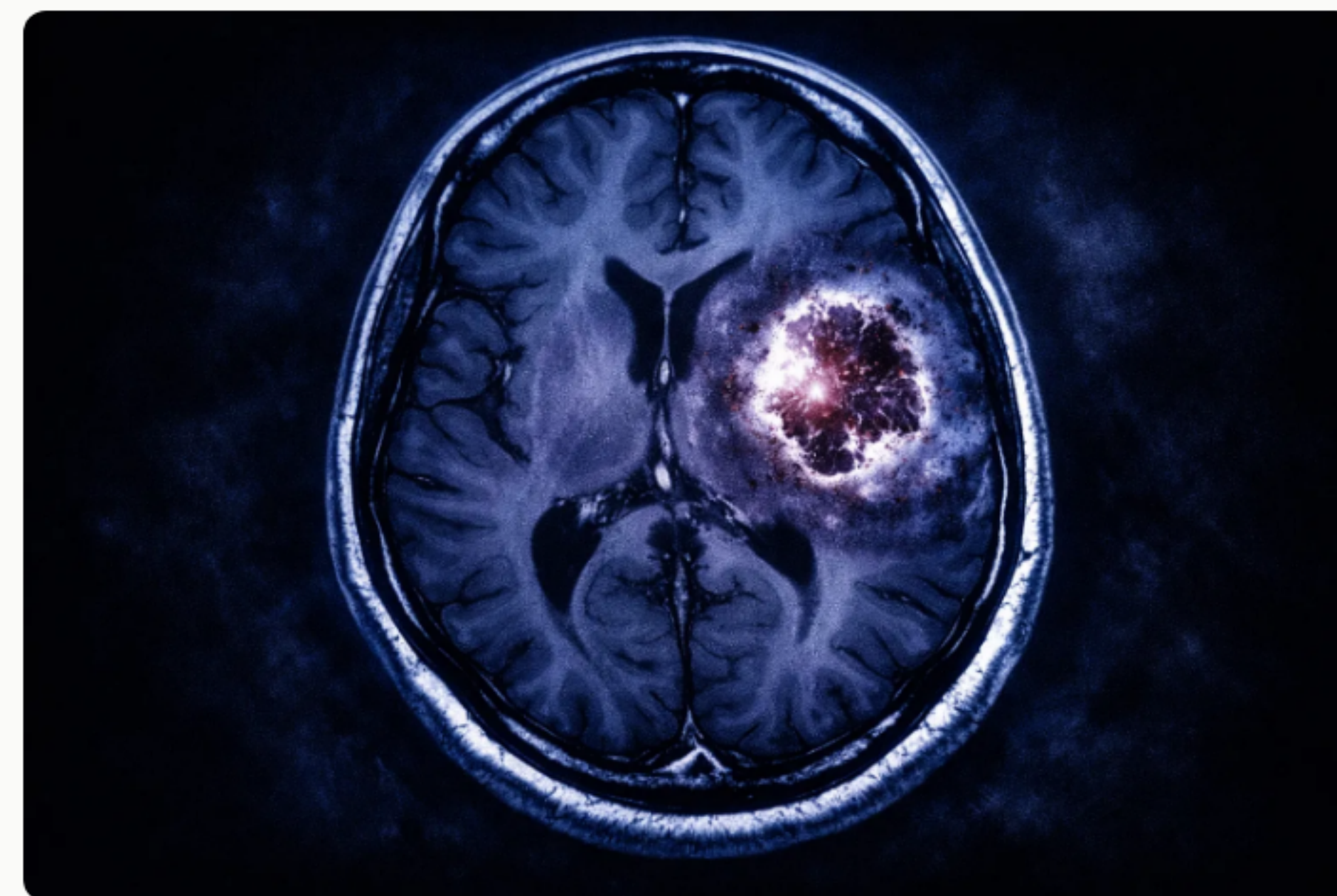
MIMUC 2026

② The Paradox

What if a feature knows too much?

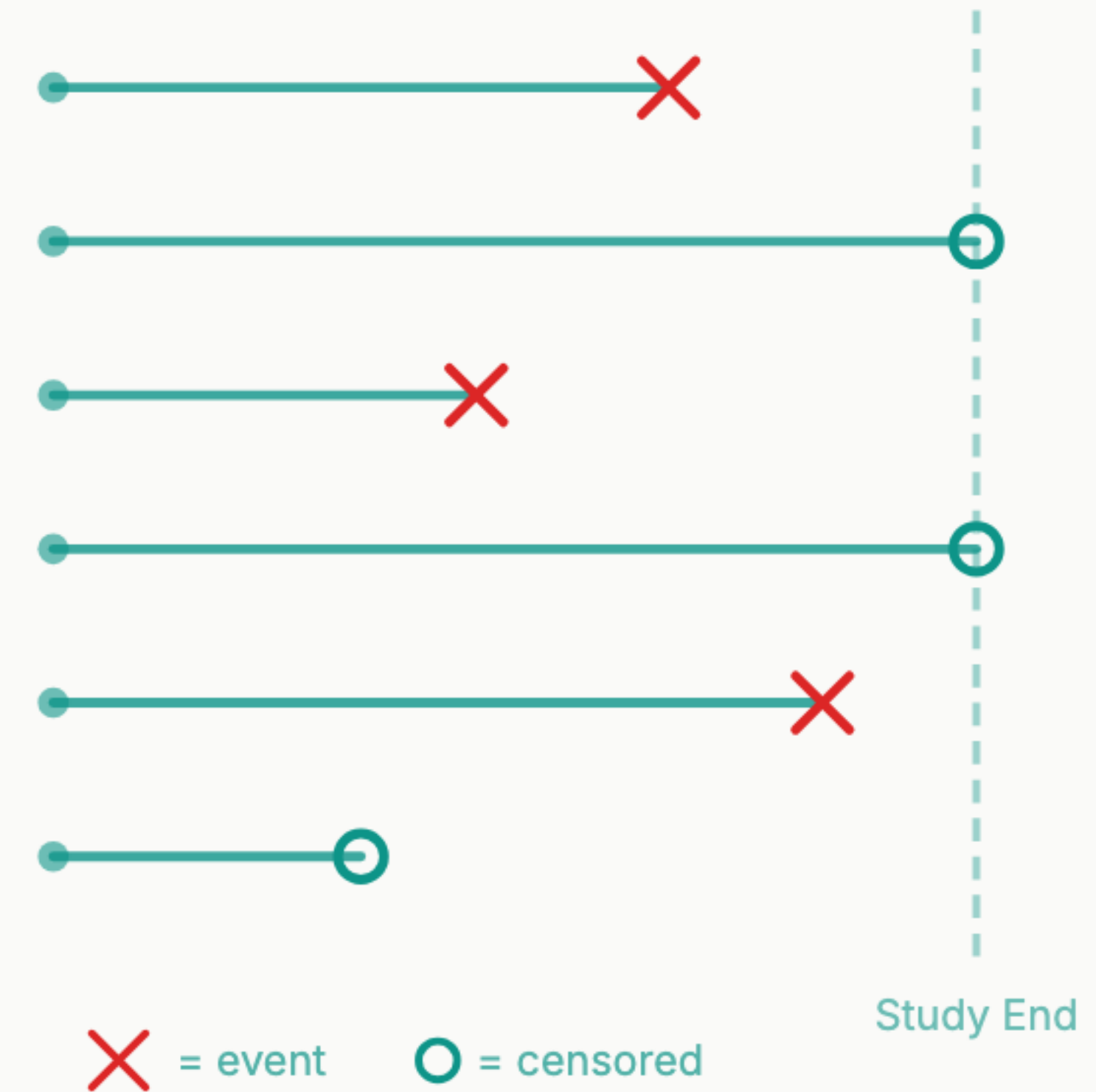
- One feature correlates with death at $r = 0.89$.
- A statistician celebrates. A mathematician is suspicious.
- But a mathematician pauses: how could any single feature 'know' this much?

This talk explores what happens when you ask that question seriously.



Why Standard Regression Fails

- Not every patient dies during observation. Some are right-censored: alive at last follow-up.
- Standard regression treats censored data as exact, introducing systematic bias.



The Elegance of Hazard Functions

One Clock, Many Speeds

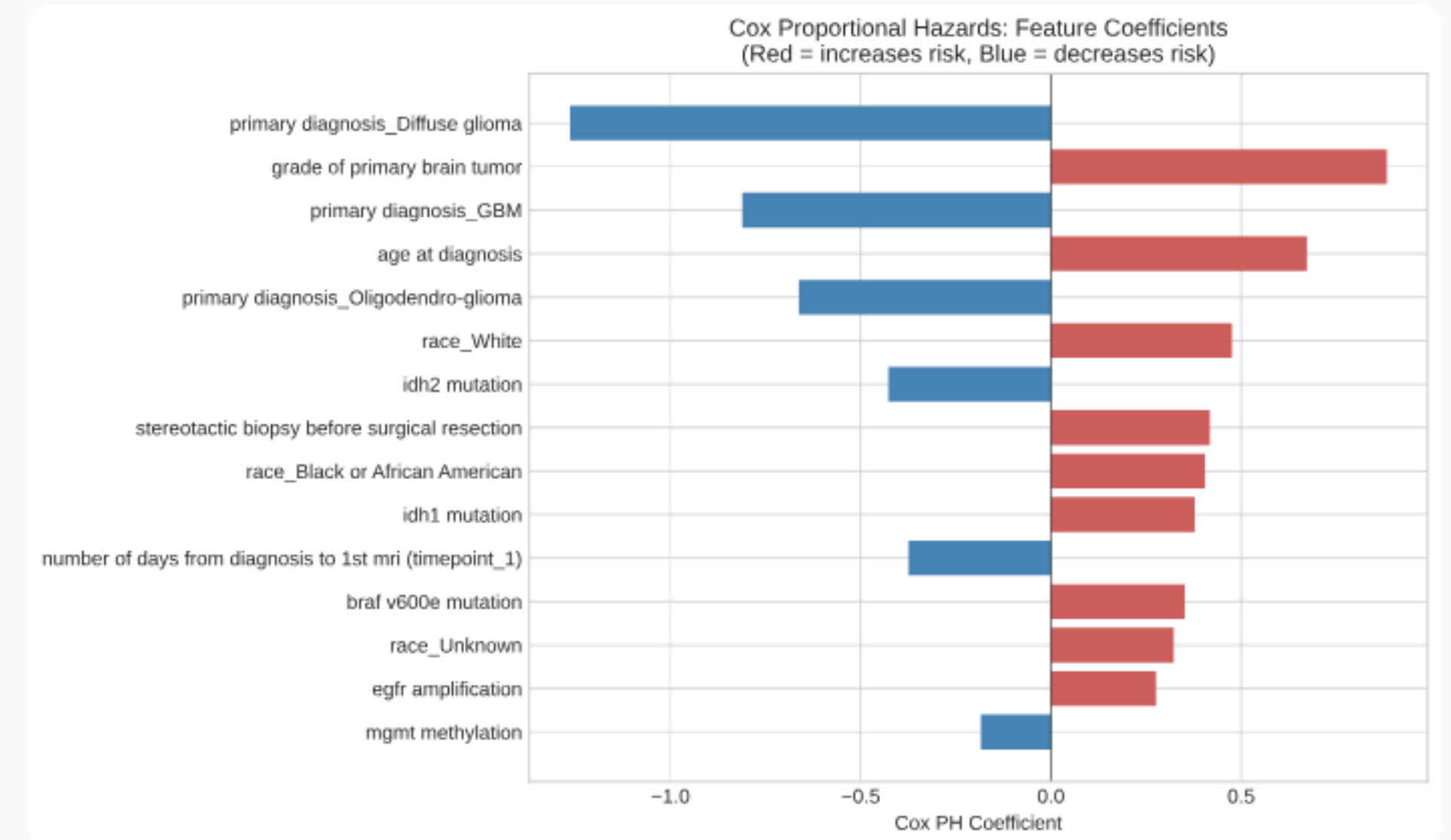
$$h(t) = h_0(t) \cdot \exp(\beta^T \mathbf{X})$$

$h(t)$ = hazard (instantaneous risk of the event)

$h_0(t)$ = baseline (the average patient's clock) β = multipliers (Age: 1.03x, IDH1: 0.34x)

$\mathbf{X}$ = patient profile (the knobs that adjust risk)

- Our Cox model: Age (**HR 1.03**, $p < 0.001$) and **IDH1 (HR 0.34)** dominate.
- Semi-parametric: no assumption on baseline shape, estimated via partial likelihood.



Measuring Truth: The Concordance Index

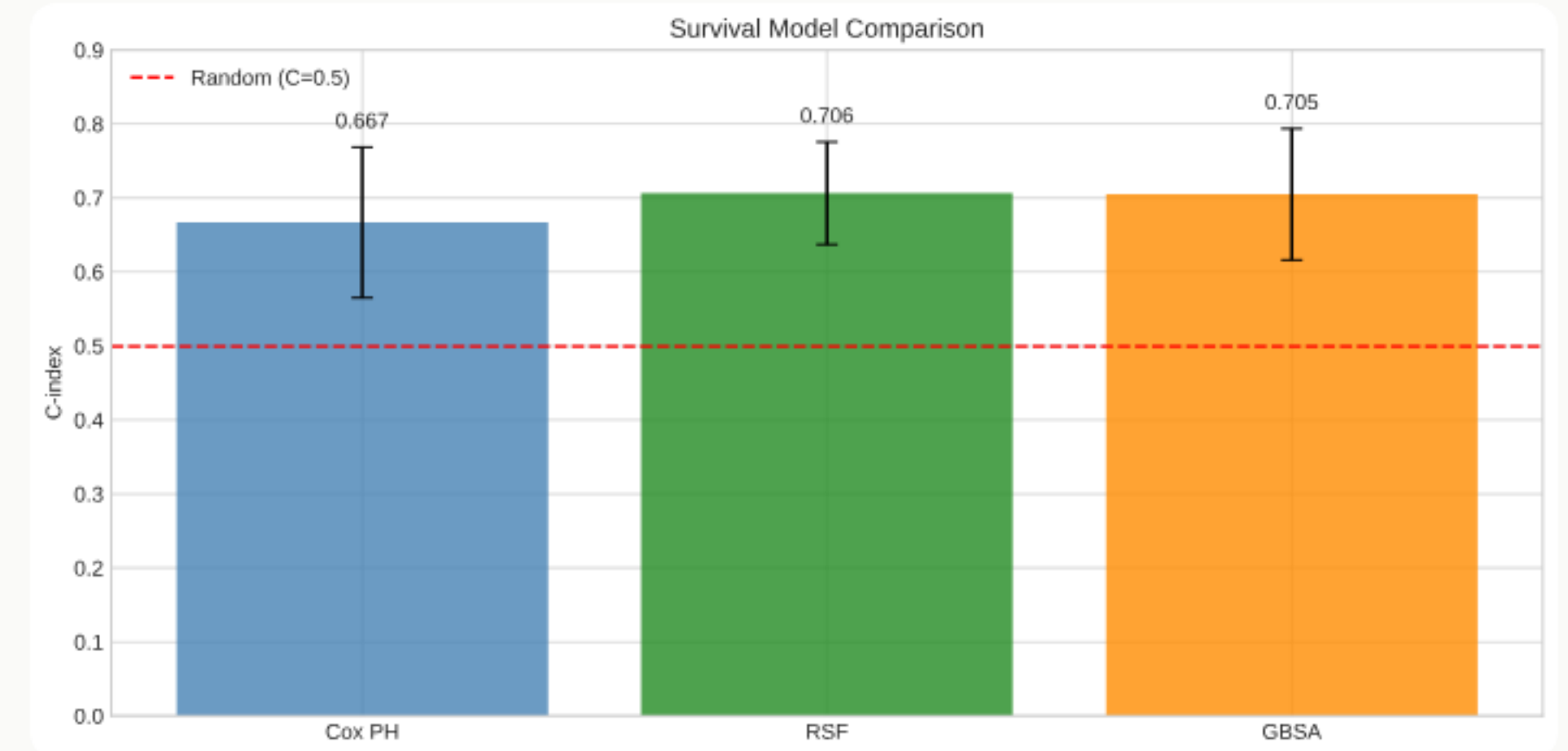
Not Accuracy, Ranking

$$C = P(\hat{r}_i > \hat{r}_j \mid T_i < T_j)$$

$\hat{r}$ = predicted risk (model's danger ranking) T = actual survival (ground truth)

C = concordance (% of patient pairs ranked correctly)

- All three models cluster near $C = 0.69$ — within 2% of each other.
- With $n = 203$, sample size is the bottleneck, not the algorithm.



🕒 The Hidden Trap: Temporal Validity

When Correlation Encodes the Answer

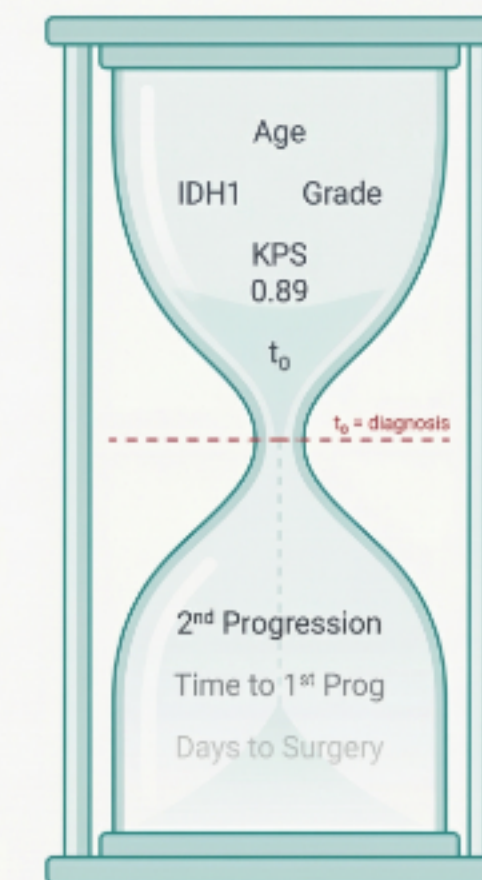
$$x_j \text{ is valid} \iff x_j \in \mathcal{F}_{t_0}$$

x_j = a feature (one column in the dataset)

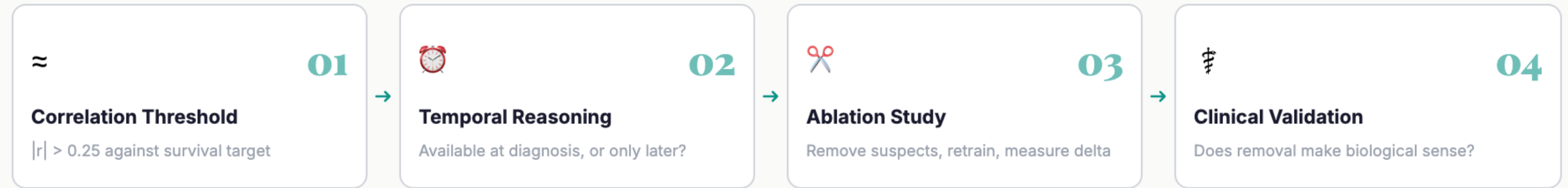
$\mathcal{F}_{t_0}$ = natural filtration (info accumulated up to diagnosis)

t_0 = diagnosis day (the prediction deadline)

- Features from $t > t_0$ observe the future, not predict it.
- "Second Progression" ($r = 0.89$): only recorded after prolonged survival.



Building a Mathematical Filter



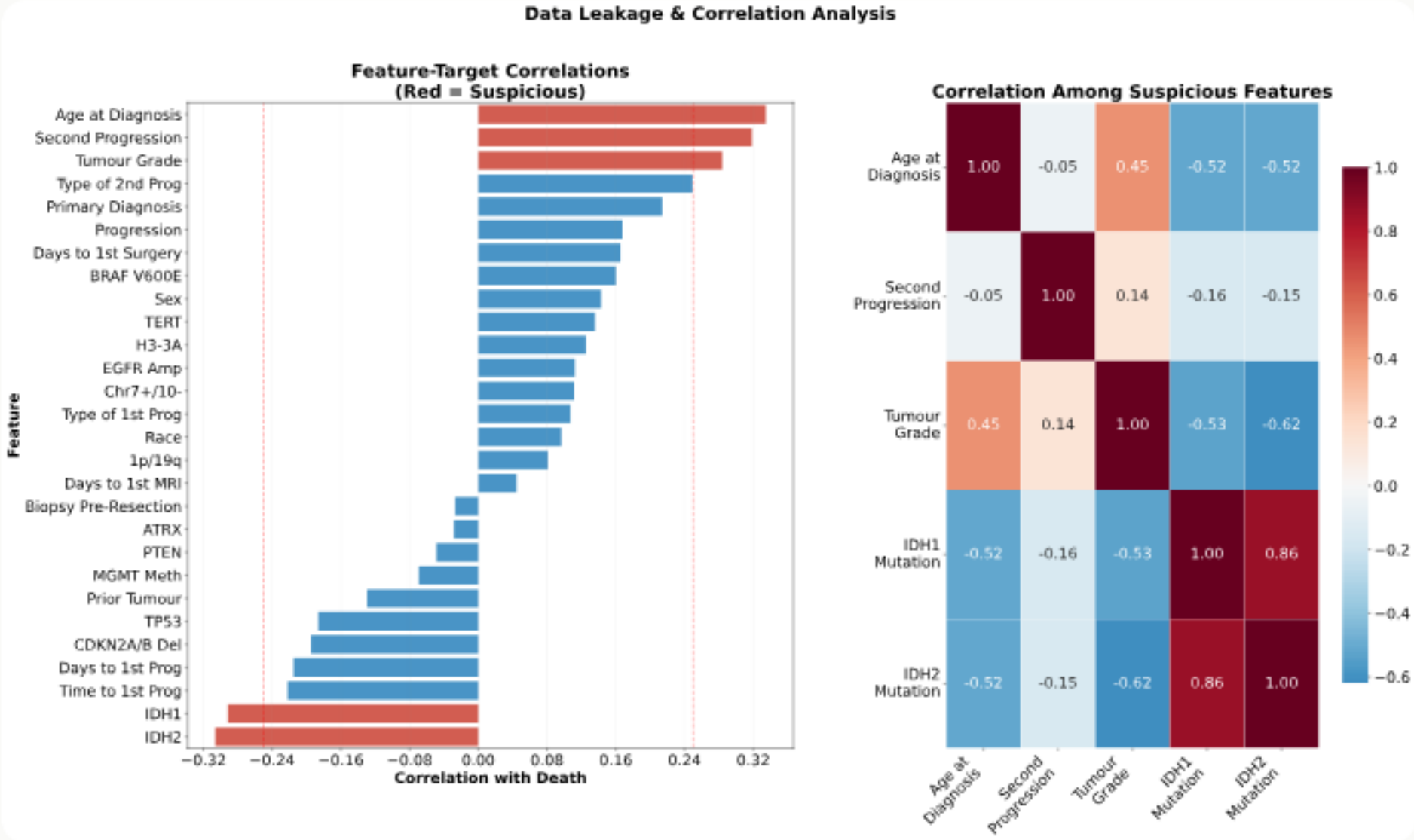
28 → **21** features

7 temporally invalid variables removed (25% reduction)

7 Features That Saw the Future

Caught Looking Forward

- **Second Progression** ($r = 0.89$): only knowable after follow-up.
- **Time to First Progression**: encodes survival directly.
- **Days to First Surgery**: reflects post-diagnosis urgency.
- **Progression** (binary): requires longitudinal observation.
- Type of 1st/2nd Progression: only available retrospectively.



The Cost of Truth

What We Lost Was Never Real

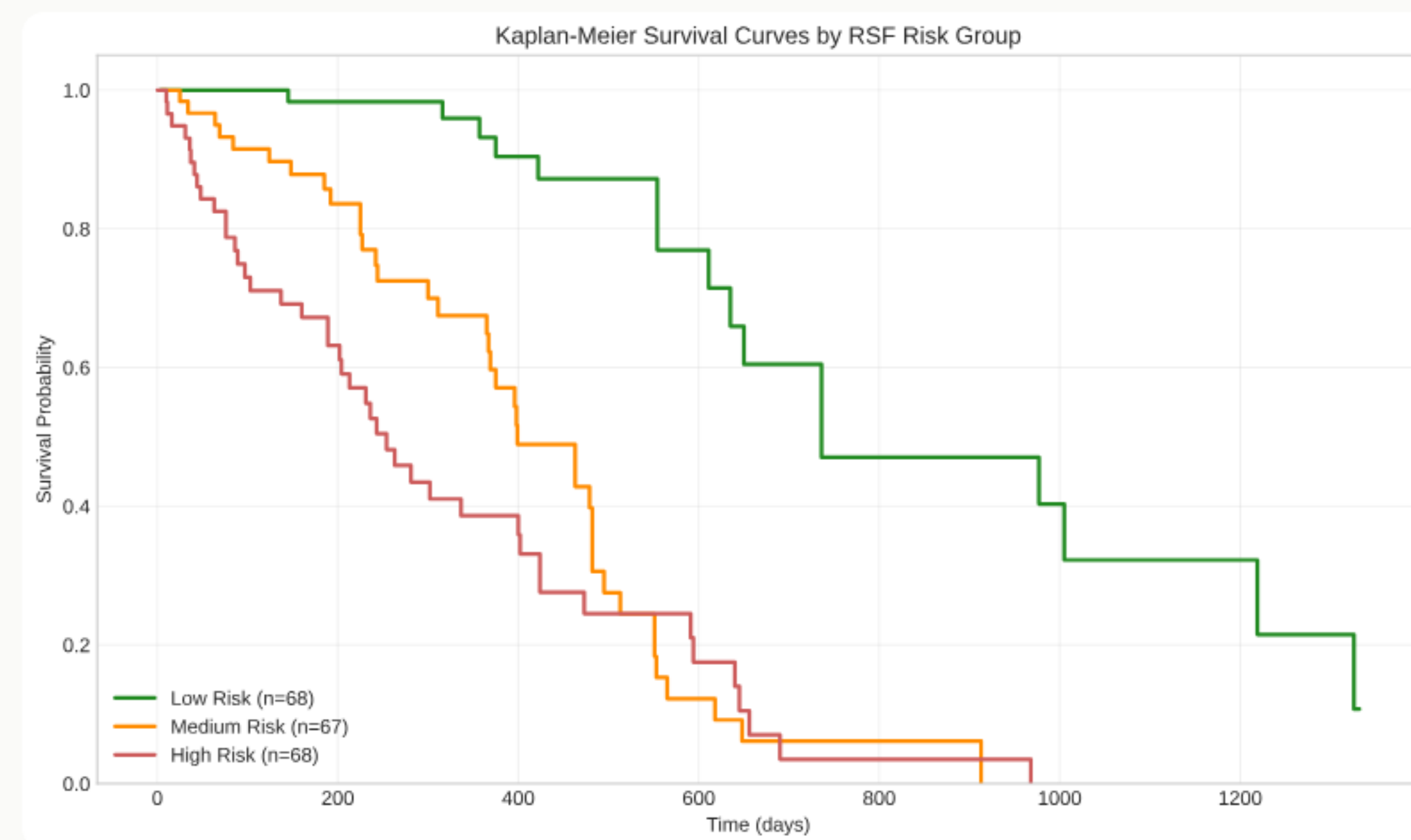
WITH LEAKAGE

0.730

HONEST

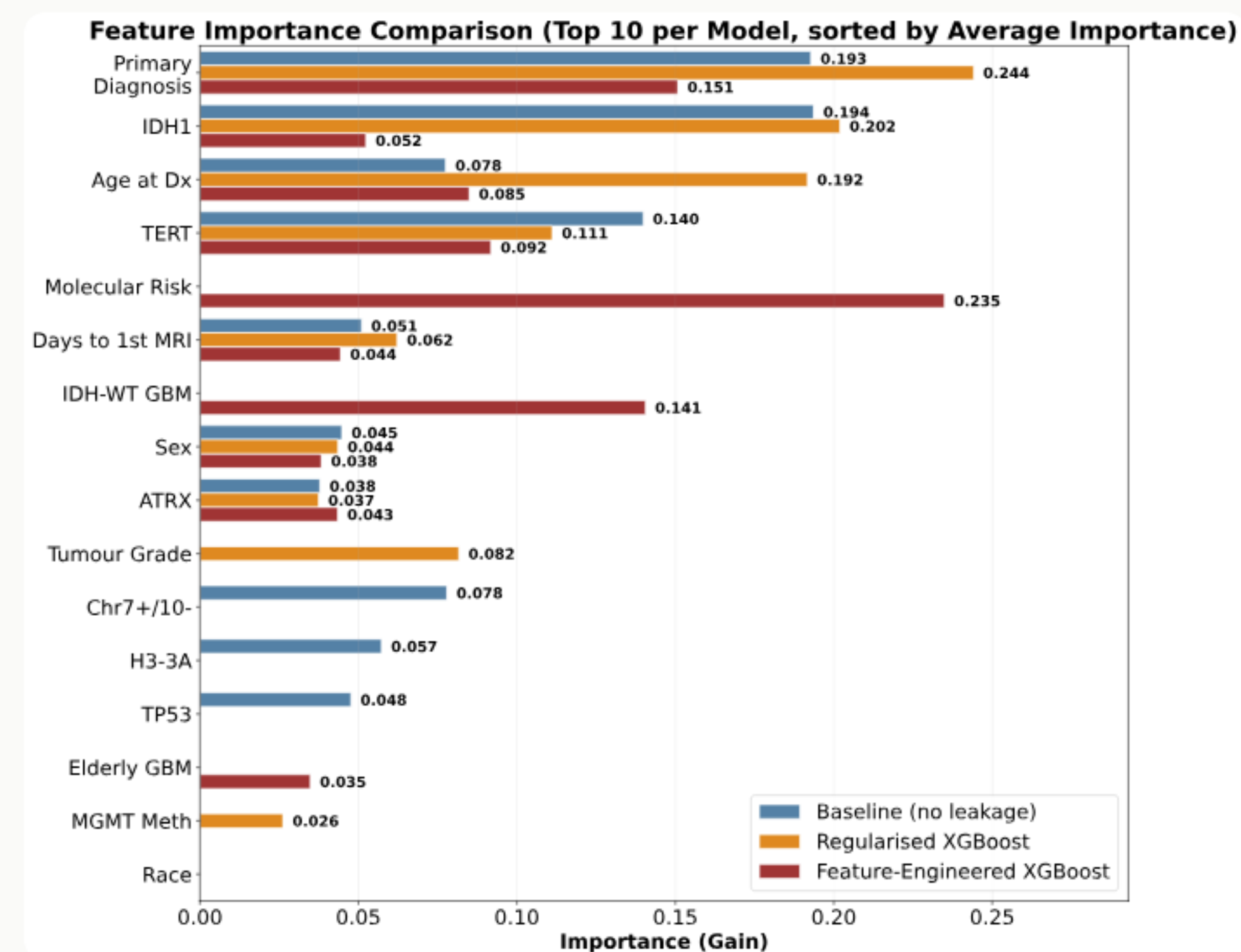
0.674

Honest AUC of **0.674** matches validated clinical tools (EORTC nomograms, C-index ~0.66). The lower number is the **real** one.



A Straight Line Suffices

- **Logistic Regression:** AUC **0.692 ± 0.119**.
- **Clean XGBoost:** AUC **0.674 ± 0.111**.
- With **n = 203**, all 5 clean models cluster in AUC **0.62–0.69**.
- **IDH1 mutation status:** **19.4%** of XGBoost feature importance.



⚡ The Principle

What the Data Actually Knows

SURVIVAL

0.67–0.72 AUC

Matches EORTC benchmarks

IMAGING

0.598 AUC

Near random chance

VOLUME

$R^2 < 0$

Cannot predict growth

Thank You!

Questions?